Literature status: separations among diameter 2 properties

Run `fa_banach_001`; source arXiv:1304.7068

Source question

Abrahamsen, Lima, and Nygaard's paper *Remarks on diameter 2 properties*, arXiv:1304.7068, defines in Definition 1.1 three properties of a Banach space X: the local diameter 2 property, where every slice of B_X has diameter 2; the diameter 2 property, where every non-empty relatively weakly open subset of B_X has diameter 2; and the strong diameter 2 property, where every finite convex combination of slices of B_X has diameter 2.

Immediately after the definition, the authors say that they do not know whether any of the reverse implications hold. In Section 4, "Some concluding remarks and questions", they again say that they do not know whether the three diameter 2 properties are really different and conjecture that they are not equal. Item (c) asks whether Theorem 3.2, the stability of the local and ordinary diameter 2 properties under ℓ_p-sums for all $1 \leq p \leq \infty$, is true for the strong diameter 2 property.

Supporting answers

Becerra Guerrero, Lopez-Perez, and Rueda Zoca's paper *Big slices versus big relatively weakly open subsets in Banach spaces*, arXiv:1304.4397, gives the first separation. The introduction says that the paper proves the existence of a Banach space satisfying the slice diameter 2 property and failing the diameter 2 property, answering negatively an open problem stated in Abrahamsen–Lima–Nygaard. In that paper's notation, "SD2P" means the slice diameter 2 property, i.e. LD2P, not the strong diameter 2 property. The decisive result is Theorem 2.4 and Corollary 2.5: every Banach space containing an isomorphic copy of c_0 admits an equivalent norm for which every slice of the new unit ball has diameter 2, while there are non-empty relatively weakly open subsets of the new unit ball with arbitrarily small diameter. Thus LD2P does not imply D2P.

The same supporting paper records the second separation: the strong diameter 2 property is strictly stronger than D2P. It cites the theorem of Acosta, Becerra Guerrero, and Lopez-Perez that

$$c_0 \oplus_2 c_0$$

has D2P but fails the strong diameter 2 property. This also answers the literal form of source item (c) negatively, since c_0 has the strong diameter 2 property but its ℓ_2-sum with itself fails strong D2P.

For the broader stability landscape, Haller, Langemets, and Nadel's paper *Stability of average roughness, octahedrality, and strong diameter two properties of Banach spaces with respect to absolute sums*, arXiv:1702.03140, gives a full absolute-sum criterion. Their Theorem 3.3 states that if X and Y have the strong diameter 2 property and N is an absolute normalized norm on $\mathbb{R}^2$, then $X \oplus_N Y$ has the strong diameter 2 property if and only if $(\mathbb{R}^2, N)$ has the positive strong diameter 2 property.

Status

The main separation question from arXiv:1304.7068 is already answered in the literature:

$$\text{SD2P} \Longrightarrow \text{D2P} \Longrightarrow \text{LD2P},$$

but neither reverse implication holds. The LD2P-versus-D2P separation is explicitly presented in arXiv:1304.4397 as answering the Abrahamsen–Lima–Nygaard open problem. The D2P-versus-strong-D2P separation is recorded there via the example $c_0 \oplus_2 c_0$, and the later absolute-sum theorem of Haller–Langemets–Nadel explains the general strong-D2P stability criterion.

This is an already-known literature answer, not a new result from the run. The packet does not answer the remaining concluding questions in arXiv:1304.7068, namely items (a), (b), (d), and (e), except for the overlap with absolute-sum stability described above.

References

- T. A. Abrahamsen, V. Lima, O. Nygaard, *Remarks on diameter 2 properties*, arXiv:1304.7068; J. Convex Anal. 20 (2013), 439–452.
- J. Becerra Guerrero, G. Lopez-Perez, A. Rueda Zoca, *Big slices versus big relatively weakly open subsets in Banach spaces*, arXiv:1304.4397; J. Math. Anal. Appl. 428 (2015), 855–865.
- R. Haller, J. Langemets, R. Nadel, *Stability of average roughness, octahedrality, and strong diameter two properties of Banach spaces with respect to absolute sums*, arXiv:1702.03140.